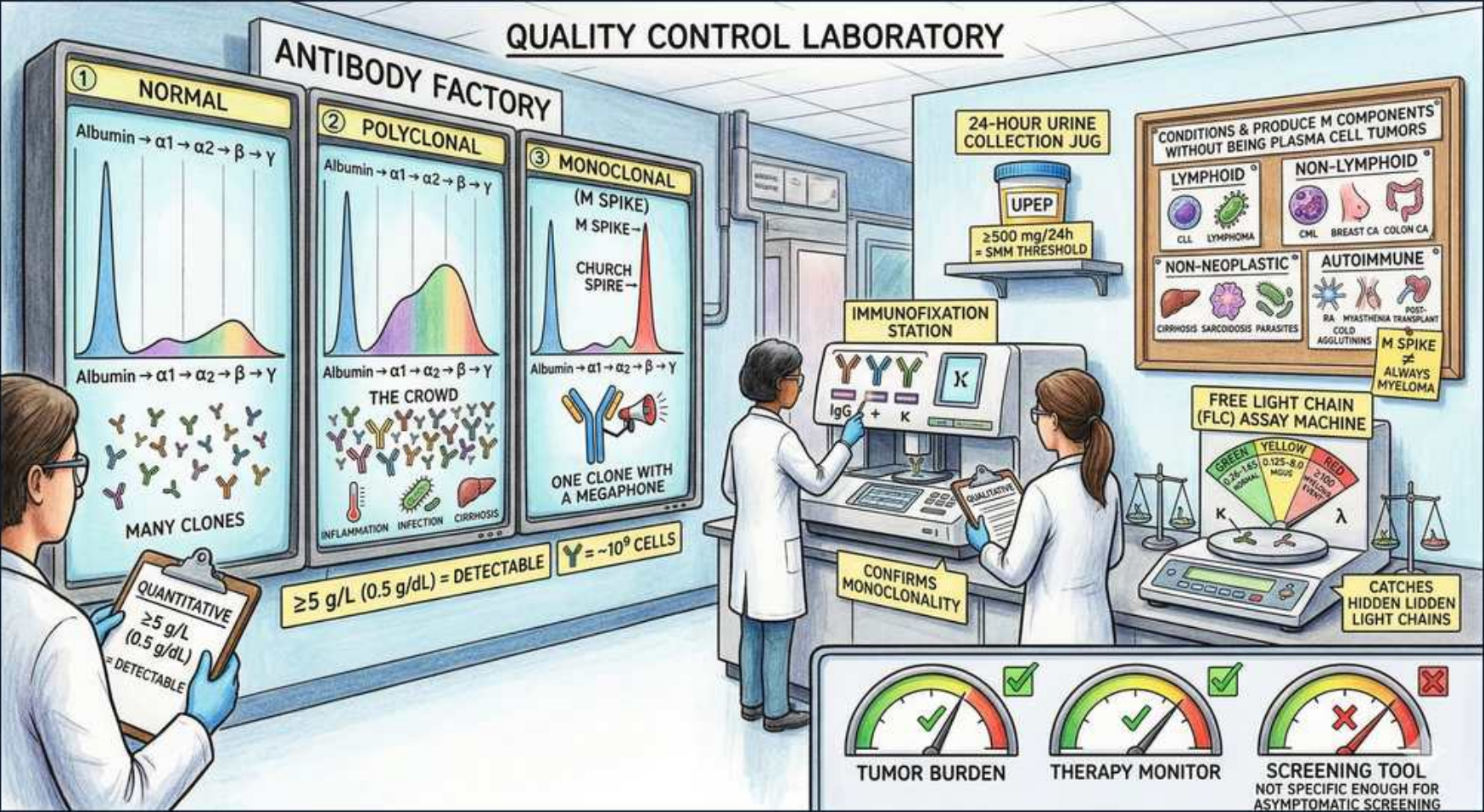


Plasma Cell — M-Spike & Quality Control Lab (SPEP/UPEP/FLC)



1. Normal SPEP

WHAT YOU SEE

Panel 1 NORMAL with smooth low broad gamma curve

WHAT IT ENCODES

Normal serum protein electrophoresis separates proteins by charge/size into albumin $\rightarrow \alpha_1 \rightarrow \alpha_2 \rightarrow \beta \rightarrow \gamma$; the gamma region (where immunoglobulins live) is a low, broad, polyclonal smear with NO discrete peak. The tall left peak is albumin. Pattern recognition: normal = smooth rolling hills, no church spire.

BOARD TRAP

WRONG answer: calling any visible gamma activity abnormal. The discriminator is shape — a normal/reactive gamma region is broad and low (many clones); only a narrow, tall, discrete peak (monoclonal spike) suggests a plasma cell dyscrasia.

2. Polyclonal gammopathy

WHAT YOU SEE

Panel 2 POLYCLONAL broad rounded gamma hump labeled THE CROWD

WHAT IT ENCODES

A polyclonal gammopathy is a broad-based, rounded rise across the entire gamma region produced by MANY different B-cell clones — a reactive, benign response. Causes include chronic infection, chronic inflammation, autoimmune disease, and chronic liver disease (often with beta-gamma bridging in cirrhosis). Mnemonic: 'the crowd' = many voices, no single megaphone.

BOARD TRAP

WRONG answer: working up a broad polyclonal hump as possible myeloma. The discriminator is that polyclonal = reactive and benign — only a NARROW monoclonal spike (single clone) raises concern for a monoclonal gammopathy/myeloma; a broad rise points to infection/inflammation/liver disease instead.

3. Monoclonal M-spike

WHAT YOU SEE

Panel 3 MONOCLONAL tall narrow church-spire peak

WHAT IT ENCODES

An M-spike is a single, sharp, narrow 'church-spire' peak in the gamma (occasionally beta) region from ONE clone making one identical immunoglobulin (the M-protein/paraprotein). It is the electrophoretic hallmark of monoclonal gammopathy — MGUS, smoldering, or multiple myeloma. SPEP quantifies the height of this spike to track disease.

BOARD TRAP

WRONG answer: equating any M-spike with multiple myeloma. The discriminator is that the spike itself only proves monoclonality — MGUS (<3 g/dL, <10% plasma cells, no CRAB) is far more common; staging requires marrow %, end-organ assessment (CRAB), and SLiM biomarkers.

4. One clone with a megaphone

WHAT YOU SEE

single antibody with megaphone under monoclonal panel

WHAT IT ENCODES

The spike represents a single malignant/clonal plasma cell line overproducing one identical immunoglobulin with one heavy chain and one light chain — hence 'monoclonal.' This is why immunofixation can pin down a single isotype (e.g., IgG-kappa). Mnemonic: one clone, one megaphone, one note shouted loudly.

BOARD TRAP

WRONG answer: expecting a normal total protein to exclude a clone. The discriminator is that the clone may be small or light-chain-only — total protein can be normal while a discrete monoclonal band/abnormal FLC ratio is present; you must look at SPEP pattern + immunofixation + FLC, not just total protein.

5. M-spike thresholds

WHAT YOU SEE

text greater than or equal to 5 g/L (0.5 g/dL) detectable, $\sim 10^9$ cells

WHAT IT ENCODES

SPEP can resolve a monoclonal band once it reaches roughly ≥ 0.5 g/dL (≥ 5 g/L), corresponding to a sizable clone (on the order of $\sim 10^9$ cells). M-protein concentration is a surrogate for tumor burden; ~ 3 g/dL serum (or $\geq 10\%$ marrow plasma cells) marks the smoldering/myeloma threshold. Numbers anchor: detection ~ 0.5 g/dL, smoldering ~ 3 g/dL.

BOARD TRAP

WRONG answer: using a small M-spike size to reassure that there is no significant disease. The discriminator is that even low-level or undetectable spikes occur in light-chain/nonsecretory myeloma — quantify the spike but always pair SPEP with serum FLC and clinical/marrow assessment.

6. 24-hour UPEP urine jug

WHAT YOU SEE

UPEP 24-HOUR URINE COLLECTION JUG with 500 mg/24h threshold

WHAT IT ENCODES

Urine protein electrophoresis on a 24-hour collection detects monoclonal free light chains filtered into urine — Bence-Jones protein. A urinary M-protein ≥ 500 mg/24h is a quantitative marker used in staging (it can substitute for the M-protein criterion in smoldering myeloma). Mnemonic: the jug catches the light chains the kidney lets through.

BOARD TRAP

WRONG answer: ordering a random/spot urine protein or relying on dipstick to detect Bence-Jones protein. The discriminator is that you need a 24-HOUR UPEP with immunofixation — a spot urine and especially the dipstick (which reads albumin only) will miss monoclonal free light chains.

7. Immunofixation station

WHAT YOU SEE

IMMUNOFIXATION STATION technician at machine

WHAT IT ENCODES

Immunofixation electrophoresis (IFE) is the confirmatory test: it overlays antisera to identify the exact heavy chain (IgG/IgA/IgM) and light chain (kappa/lambda) of the monoclonal protein, and is more sensitive than SPEP for small bands. SPEP/UPEP quantify; IFE confirms and TYPES. Mnemonic: 'fix the type' = immunofixation names the clone (e.g., IgG- κ).

BOARD TRAP

WRONG answer: using SPEP alone to diagnose and characterize the M-protein. The discriminator is that SPEP only quantifies a suspicious peak — immunofixation is required to CONFIRM monoclonality and assign the heavy/light chain isotype, and can detect bands too small for SPEP.

8. Free light chain (FLC) assay

WHAT YOU SEE

FREE LIGHT CHAIN (FLC) ASSAY MACHINE scale

WHAT IT ENCODES

The serum free light chain assay measures unbound kappa and lambda and their ratio (normal ~0.26–1.65); an abnormal ratio indicates a clonal light-chain excess. It detects light-chain-only and nonsecretory disease that SPEP/UPEP miss, and an involved:uninvolved FLC ratio ≥ 100 is a SLiM myeloma-defining biomarker. Mnemonic: the scale tips κ vs λ — imbalance = clone.

BOARD TRAP

WRONG answer: excluding a plasma cell disorder because SPEP/UPEP are negative. The discriminator is that light-chain-only / nonsecretory myeloma produces no intact-Ig spike — order serum FLC; an abnormal ratio (or ≥ 100) reveals (and can even define) the disease.

9. Conditions producing M-protein

WHAT YOU SEE

box listing lymphoid, non-lymphoid, autoimmune, non-neoplastic causes

WHAT IT ENCODES

M-proteins are not specific to myeloma: they appear in plasma cell disorders (MGUS, myeloma, Waldenström, amyloidosis), other lymphoid malignancies (CLL, lymphoma), chronic infections, and autoimmune/non-neoplastic states. The detected isotype hints at the source — IgM $\rightarrow$ Waldenström/lymphoma; IgG/IgA $\rightarrow$ MGUS/myeloma. Context plus isotype directs the workup.

BOARD TRAP

WRONG answer: diagnosing multiple myeloma from the presence of an M-protein alone. The discriminator is the broad differential — an IgM spike especially points toward Waldenström/lymphoma rather than myeloma, and many M-proteins are benign MGUS; correlate isotype, marrow, and end-organ findings.

10. Tumor burden gauge

WHAT YOU SEE

TUMOR BURDEN green-yellow-red dial

WHAT IT ENCODES

M-protein concentration correlates with plasma cell tumor burden and feeds prognostic staging — though modern myeloma staging (R-ISS) actually uses beta-2-microglobulin, albumin, LDH, and high-risk cytogenetics rather than the spike size itself. Higher M-protein generally signals greater disease bulk. Mnemonic: bigger spike, bigger clone.

BOARD TRAP

WRONG answer: staging myeloma severity by M-spike height. The discriminator is that prognostic staging (R-ISS) relies on beta-2-microglobulin, albumin, LDH, and cytogenetics — the M-protein tracks burden and response but is not the staging system.

11. Therapy monitor gauge

WHAT YOU SEE

THERAPY MONITOR dial

WHAT IT ENCODES

Serial M-protein measurement (SPEP/UPEP/FLC) is the principal tool to monitor treatment response and detect relapse; response is defined by percent reduction in the monoclonal protein (and FLC for light-chain disease). Always follow the SAME marker the patient secretes. Mnemonic: track the same spike you started with.

BOARD TRAP

WRONG answer: monitoring a light-chain or nonsecretory myeloma patient with SPEP because that's standard. The discriminator is matching the assay to the secreted protein — follow serum FLC (and 24-h urine) in light-chain/nonsecretory disease, since SPEP won't reflect their response.

12. Screening tool gauge

WHAT YOU SEE

SCREENING TOOL dial NOT SPECIFIC FOR ASYMPTOMATIC SCREENING

WHAT IT ENCODES

SPEP is a diagnostic test for SYMPTOMATIC/suspected patients, not a population screening test — MGUS is common (~3% over 50) and overwhelmingly benign, so screening asymptomatic people causes overdiagnosis and anxiety. Order SPEP/UPEP/FLC when clinical features suggest a dyscrasia (unexplained anemia, renal failure, hypercalcemia, lytic lesions, neuropathy). Mnemonic: test the symptomatic, not the population.

BOARD TRAP

WRONG answer: adding SPEP to a routine health check or working up every incidental MGUS aggressively. The discriminator is the low yield/high false-alarm rate of asymptomatic screening — reserve SPEP for patients with features suggesting a plasma cell disorder.